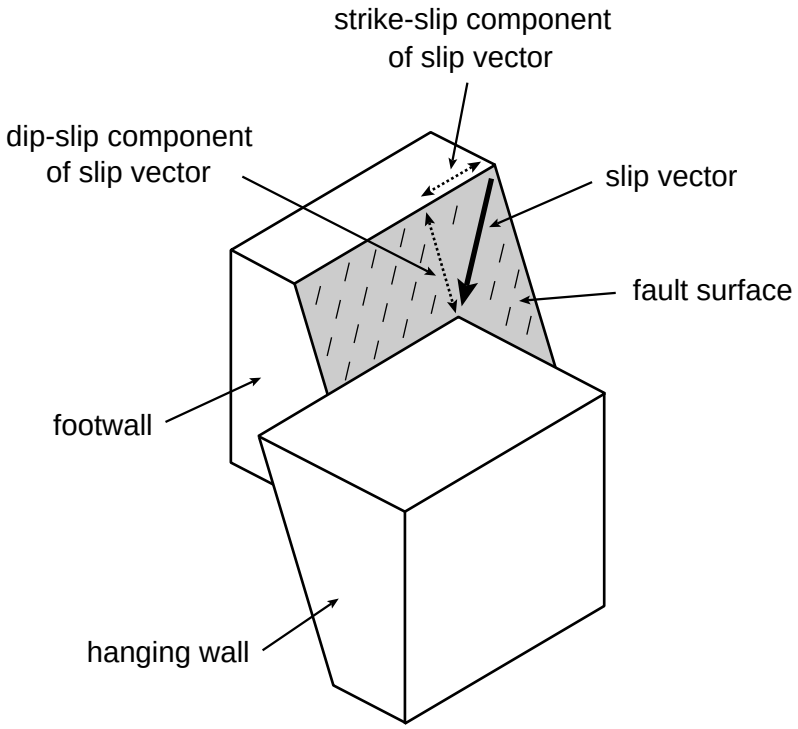
THE GEOMETRY OF FAULTS (1)

A fault is a fracture in a rock mass along which there has been observable displacement. In order to describe the geometry of a fault fully it is necessary to specify

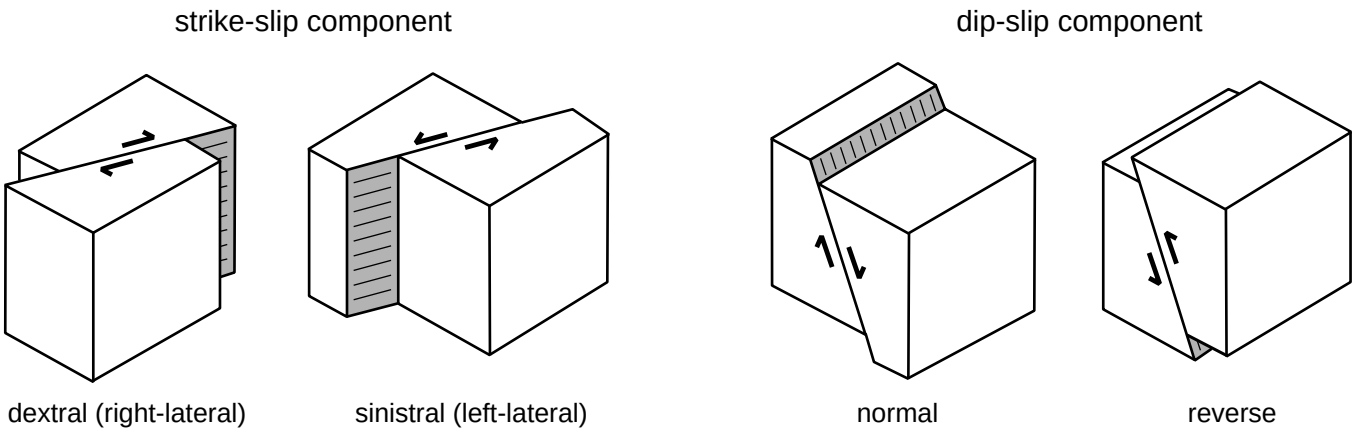
(a) the orientation of the fault surface, and

(b) the orientation and magnitude of the slip vector, that is, the line connecting points on either side of the fault surface which were originally coincident

(c) the sense of movement along the slip vector



- **The orientation of the fault surface** is specified by the strike/dip (or dip/dip direction) of that surface
- **The orientation of the slip vector** is specified by its plunge/plunge direction (or its pitch in the fault surface)
- **The magnitude of the slip vector** may be specified either by the magnitude of the vector itself, or by the magnitude of its strike-slip and dip-slip components (or preferably, by both methods)
- **The sense of movement along the slip vector** is given by the relative displacement of the units on either side of the fault surface and must be specified for both the strike-slip and dip-slip components:



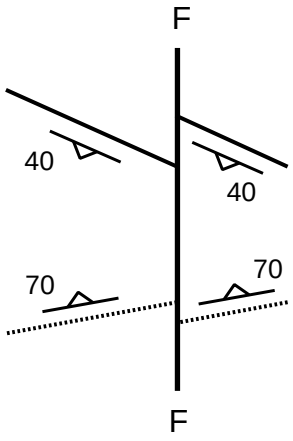
THE GEOMETRY OF FAULTS (2)

Determining the orientation of the fault surface from a map

• Obtained by drawing stratum contours on the fault.

Assessing the sense of movement on a fault shown on a map

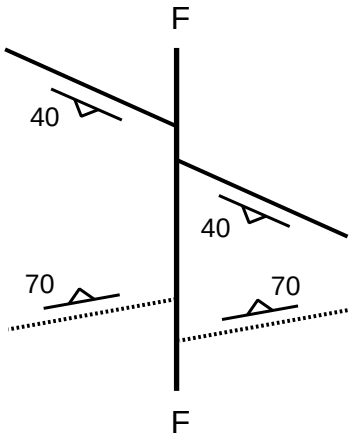
• Obtained by examining the offset of contacts along the fault. Note that the offset of a contact along a fault shown on a map does not imply that there was a strike-slip component to the slip vector − examine the accompanying handout where several different examples of apparent strike-slip offsets produced from purely dip-slip faulting are shown.



Two differently oriented contacts
having two different apparent offsets

↓

dip-slip component to the slip vector



Two differently oriented contacts
with the same apparent offsets

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no dip-slip component to the slip vector

• Determining the sense of movement (normal or reverse and dextral or sinistral) inevitably requires a mental picture of what direction of movement could produce the observed apparent offsets. Such thought is often more simply carried once the orientation of the slip vector has been established.

Quantifying the magnitude and orientation of the slip vector

• In the situation where the fault is a pure ***dip-slip fault*** and the strike of the offset contacts is close to that of the fault itself, the magnitude of the slip vector may be obtained by drawing a vertical cross-section parallel to the dip direction of the fault. In this case the orientation of the slip vector is the same as the dip/dip direction of the fault.

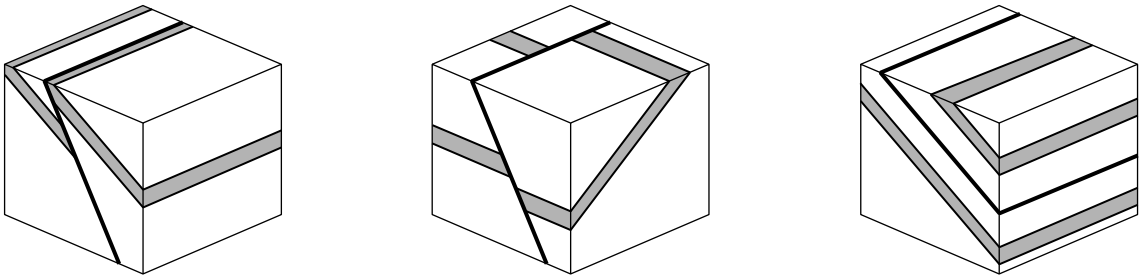
• In the situation where the fault is a pure ***strike-slip fault***, the magnitude of the slip vector is simply the magnitude of the apparent offset (making due allowance for the effect of topography on the map position of the contacts along the fault). In this case the orientation of the slip vector is the same as the strike of the fault.

• In the general case (***oblique-slip faults*** and other dip-slip faults) the magnitude and orientation of the slip vector must be obtained by drawing a fault plane (juxtaposition) section, that is, a cross-section drawn in the plane of the fault showing both the hanging wall and the footwall on the same section.

THE GEOMETRY OF FAULTS (3)

Some redundant terminology used to describe the geometry of faults

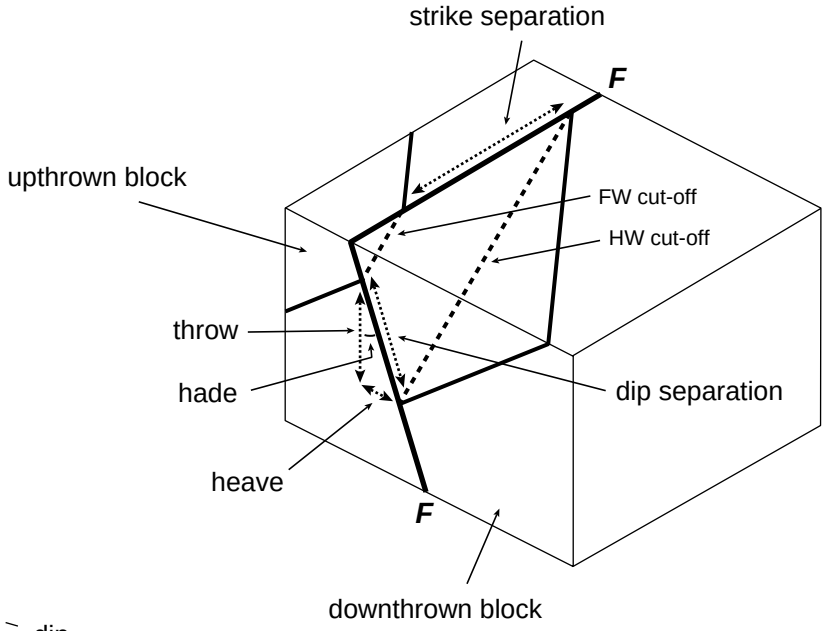
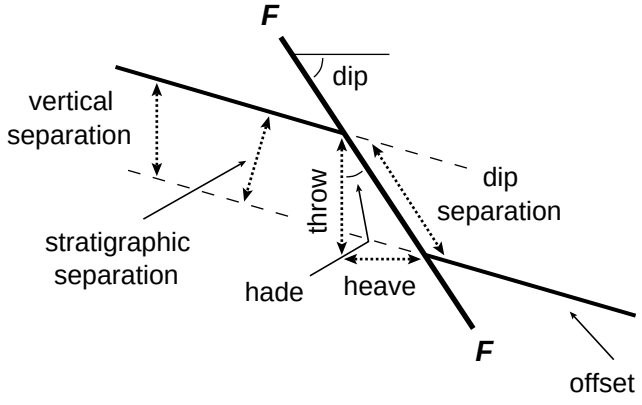
• Faults are described by some with reference to their orientation relative to the structure of the beds it displaces. Thus a ***strike fault*** is one which has a strike parallel to the strike of the beds it displaces, a ***dip fault*** is one which strikes parallel to the dip direction of the beds it displaces, and a ***bedding plane fault*** is one which is parallel to the bedding (and hence a special case of strike fault).



strike fault | dip fault | bedding plane fault

Separation

• If one only has the strike/dip of the fault and the apparent offset of one surface of known strike/dip (*e.g.*, some lithological contact), it is not possible to determine the slip vector. However, something may be said about the magnitude of the slip by specifying the ***separation*** across the fault. The diagram below illustrates a number of the terms used in the description of separation.

Strike separation is apparent offset; the other terms are defined in the vertical plane containing the dip direction of the fault.

Cut-off lines are the intersection of the surface of interest with the fault. There are two of these, one for the footwall and one for the hanging wall. Their map separation is heave.